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Lattice-Boltzmann Pore-Scale Simulator for Complex Multiphase Flows in the Digital Twins of Reservoir Rocks

A. Vishnyakov, Aramco Innovations, Moscow, Russia / Department Physics, Moscow State University, Moscow, Russia; Q. Sahu, Saudi Aramco EXPEC ARC, Dhahran, Saudi Arabia

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Abstract

Prediction of multi-phase flows in tight pore materials is a great importance for oil recovery by filtration, CO₂ sequestration, and acid stimulation of carbonate reservoirs. The paper presents a fast highly scalable Lattice-Boltzmann simulator of multiphase flows in the digital twins of reservoir rocks obtained by computer tomography and a validation against experimental data. First, the method is tested against single-phase permeabilities of sandstones and carbonates, including very tight Achimov formations. Then, different models of interfacial interactions are applied in simulations of concurrent multiphase flows. A novel simulation protocol is introduced: a preliminary drainage simulation is followed by a series of two-phase concurrent flow simulations, which mimic filtration. In each filtration simulation the ratio between the volumes of the two phases set at the inflow boundary and thus the saturation—permeability curves are obtained. The protocol is computationally cheaper than the current practices and provides a precise control over saturation levels. The parameterization of the LB models accounts for the ratio between the interfacial forces and inertia, which is especially important in imbibition simulations. By adding spontaneous imbibition cycles between the steady state permeability measurements, the fluid is relaxed towards its global free energy minimum, which affects the irreducible water and oil ratios. Quantitative agreement with experimental relative permeabilities is achieved. Then the simulation methodology is adopted to study the dynamics of liquid-gas system and compared with the experiment on foam drainage, with very reasonable results.

Introduction

The behavior of multiphase fluids within porous media governs processes vital to diverse fields, including enhanced hydrocarbon extraction, subsurface CO₂ storage, filtration and absorption technologies, contaminated soil cleanup, and biomedical engineering. Fluid movement arises from external fields, viscous stresses, and interfacial interactions. The macroscopic transport properties are dictated by pore-scale phenomena, which are themselves controlled by external conditions and the medium's structural features—such as pore network geometry, surface variations, and flow path complexity. Micron scale flow instabilities

like Haines jumps and snap-off events, inherent to these complex systems, necessitate computationally viable yet robust simulation methods for accurate multiphase flow prediction.

Realistic 3D pore geometry reconstruction, enabled by advances in micro-X-ray computed tomography (μ XCT), now facilitates *in silico* flow studies using direct and indirect simulation. Direct methods solve the Navier-Stokes (NS) equations explicitly, providing high-resolution velocity and temperature data at high computational cost. The computationally lightweight alternatives, primarily pore network models (PNMs), sacrifices detail and accuracy (Ryazanov, van Dijke et al., [Blunt 2001](#)). The Lattice-Boltzmann method (LBM) offers a highly scalable approach between molecular and continuum descriptions ([Shan and Chen 1993](#), [Anderson, Tannehill et al. 2020](#)), modeling flow via distribution function evolution on a lattice. Permeabilities of tight-pore rock samples obtained in LB simulations agree well with experiments and other computations ([Olhin and Vishnyakov 2023](#)). Because of lower computational expenses, LB simulations are often employed to generate databases for machine learning models ([Akeweje et al 2023](#), [Avilkin et al. 2025](#), [Vishnyakov 2025](#)). Multiphase LBM implementations utilize interfacial models like Shan-Chen (SC) ([Shan and Chen 1993](#)), free energy (FE) ([Swift et al. 1996](#)), and color gradient (CG) ([Gunstensen 1991](#)). Despite promising results ([Pan et al. 2006](#), [Chen et al. 2015](#)), quantitative experimental validation of LBM for multiphase flow in *realistically complex* porous media (contrasted with simple single-scale models) is still quite scarce.

Our objectives are: (i) developing a computationally efficient, simple, and accurate protocol for quantitative multiphase flow prediction in complex tight porous media; and (ii) quantitatively testing existing interfacial models using experimental data. Addressing these goals required the application of contemporary modeling techniques featuring advanced wetting boundary conditions and continuum surface force (CSF) methods.

Rock samples

This paper examines several rock sample groups. **Group 1:** Wide-pore sandstones from the Imperial College London (ICL) collection ([Leu et al. 2019](#)). Their 3D maps have resolutions of 2.85-10.0 $\mu\text{m}/\text{voxel}$ (coarser than other groups here) and physical sizes between 1.155 mm and 4.5 mm, mapped on grids of 300^3 to 450^3 voxels. Literature provides experimental single-phase permeability data obtained via conventional methods: measuring pressure drop during water injection ([Dong 2009](#), [Raeini et al. 2014](#)). For the Gildehauser sandstone, two-phase relative permeability data is also available ([Raeini et al. 2017](#)). **Group 2:** Includes selected lower-porosity, lower-permeability ICL samples (two sandstones, two carbonates). While experimental permeabilities are unavailable, published modeling results exist ([Muljadi 2016](#), [Raeini et al. 2017](#), [Suzuki et al. 2021](#)). Their pore structure maps feature significantly finer resolution (1000^3 voxels), enabling detailed pore network characterization. Permeability calculations used finite volume simulations as a benchmark ([Raeini et al. 2017](#)). **Groups 3 & 4:** Comprise tight sandstones from the Achimov formation. μ XCT imaging achieved a much finer spatial resolution of 1.2 $\mu\text{m}/\text{voxel}$. Although experimental data is lacking for both groups, direct simulations (Hydrodynamic density functional theory DHD ([Koroteev et al. 2014](#)), DiMP-Hydro NS solver ([Balashov et al. 2020](#))) have been reported ([Orlov et al. 2021](#)). The pore structures of select samples are shown in [Figure 1](#).

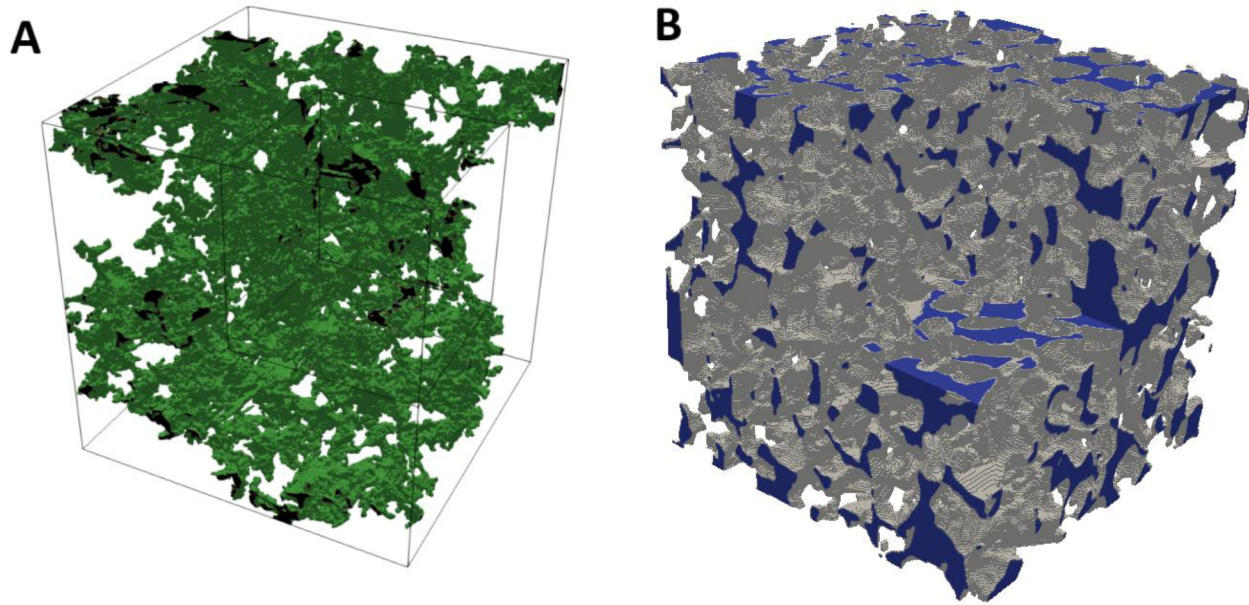


Figure 1—Visualization of the pore space of select rock samples in this study. (a) Tight-pore Achimov sandstone 3_E sample (group 3) with overall porosity of 7% (600³ lattice, 0.72 mm height) (b) Gildehauser 1_G sandstone sample with wider pores (group 1) (893×897×570 lattice, 4.4 mm in height).

LBM calculations

This study focuses on three prevalent interfacial interaction models within the Lattice Boltzmann Method (LBM) for simulating two-phase flow in porous media under both steady-state and dynamic conditions: the Color Gradient (CG) model, the Shan-Chen (SC) inter-particle potential model, and the Free Energy (FE) model. While all three are extensively validated against numerous benchmarks and share a common foundation in basic LBM dynamics, their specific implementations reveal significant operational differences.

The CG (Rothman and Keller 1988, Gunstensen, Rothman et al. 1991), employs separate distribution functions for two immiscible fluids, labeled as "Red" and "Blue." This model introduces interfacial tension through a perturbation operator based on the local color gradient $G(x, t)$ defined as:

$$G(x, t) = \sum_i e_i [\rho^R(x + e_i, t) - \rho^B(x + e_i, t)] \quad (1)$$

where $\rho(x, t)$ effective LB densities of the immiscible liquids. An advantage of the CG model is an explicit tracking of the fluid interface, which allows for a detailed representation of interfacial dynamics (Liu et al. 2016, Leclaire et al. 2017) and is particularly suitable for simulations where interface behavior significantly influences the flow. The surface tension is incorporated as a body force at the interface between the fluids $\mathbf{F} = \gamma \kappa \nabla \phi / 2$, where γ is the interfacial tension, $\kappa = \nabla(\nabla \phi / |\nabla \phi|)$, and $\nabla \phi$ is the order parameter distinguishing between the fluids. The evolution equations are modified to include the interfacial force. The Geometrical Wetting Model numerically imposes the desired contact angle θ at the three-phase contact line by modifying the fluid interface direction. The wetting boundary condition is implemented by extrapolating ϕ to the solid boundary nodes: $\phi(x_b) = \cos \theta$. This approach ensures the minimization of spurious color gradients at the fluid-solid boundary.

The SC pseudopotential (Huang, et al. 2007) model introduces microscopic interactions between neighboring particles through a mean-field force that naturally leads to a phase separation and generates interfacial tension without explicit interface tracking. The SC approach allows for large density and viscosity contrasts between fluids. On the other hand, spurious velocities near the interface are often observed at higher density ratios, rising concerns about simulation accuracy. Additionally, the interfacial tension is less

precisely controlled compared to the CG and FE models, which may become a limitation when accurate interfacial dynamics are essential.

The FE model employs a free energy functional to account for interfacial tension effects and interface dynamics in a consistent manner. It accurately describes the capillary effects and wettability, which are critical in micron-scale porous media flows where the Reynolds numbers Re (Li et al. 2016) are very low, and allows for precise implementation of wetting boundary conditions. The details were described by (Swift et al. 1996).

Simulations of the liquid-gas flows were based on the same order parameter approach, but certain specifics of the gas systems should be taken into account. The approach was taken from (Ataei et al. 2021). Since the gas density can be neglected in comparison with the liquid densities, the liquid moves freely with respect to the gas and the only influence of the gas phase is the pressure it exerts on the free surface. Therefore we involved the volume-of-fluid (VOF) method, which introduces a dimensionless scalar variable α to track the interface location:

$$\alpha = M\rho(\Delta t)^3, \quad (2)$$

where M is the liquid mass in a cell. $\alpha = 1$ in cells that belong to the liquid phase and $\alpha = 0$ in the cells that belong to the gas phase. $0 < \alpha < 1$ in the interfacial cells. The order parameter is calculated in the process of simulation based on the mass exchange between the neighboring cells. Therefore, the interface location is explicitly tracked. An interface cell becomes a liquid cell when $\alpha \geq 1$ and a gas cell if $\alpha \leq 0$. If $\alpha < 0$ or $\alpha > 1$, the excess is distributed around the neighboring interfacial cells. $\alpha(\mathbf{r})$ is employed for interface tracking.

The gas phase was assumed ideal, each bubble was characterized with the pressure; the gas density in the equations governing the streaming and collision steps were "corrected" with via the Laplace equation that accounts for γ and the local curvature. Disjoining pressure $\Pi = (\partial G / \partial h)_{N,P,T}$ was applied in the films separating the bubbles: $\Pi(h) = k_{\Pi}(h - h_c)$ at $h < h_c$, 0 at $h \geq h_c$, that is, the disjoining force is always repulsive. We assumed the magnitude of $k_{\Pi} = A$ and the cutoff $h_c = B$. Such a shape of $\Pi(h)$ might be caused by polymer coils trapped between the bubbles, but here we do not aim at any particular practical system. Originally, the repulsive disjoining pressure was introduced for the sake of numerical stability, and its effect on the foam properties is yet to be investigated. The calculation of h is not very trivial in a disordered system. Following (Ataei et al. 2019), h is evaluated by first moving along the first bubble interface normal away from the bubble, until reaching an interface that belongs to the second bubble. Then, the interfaces of each bubble are geometrically reconstructed using a piecewise linear interface calculation, and h is calculated as the distance between the reconstructed interfaces.

The gas solution in water is assumed infinitely diluted, its concentration in the liquid phase is described with a separate distribution function, and the simplified diffusion equation is solved on D3Q7 lattice topology, exploiting the diffusion linearity (Ataei et al. 2019). The gas diffusing into bubbles is calculated using the concentration gradient at the liquid-bubble interface.

Simulation protocol in multiphase flows

This work elaborates the following protocol designed to efficiently match the steady state conditions and better represent the flow regime that may be found in the process of filtration. At the first stage, a primary drainage simulation is performed to obtain a fluid configuration at the threshold of two-phase mobility. From that state, a mixture of oil and water is injected directly into the porous medium at a specified oil/water volume ratio, effectively modeling a two-phase steady state filtration. Setting the oil/water ratio in the inflow ensures a precise and direct control over the overall saturation in the system and thus allows exploring the range of saturation levels relevant to reservoir conditions.

(Boek, et al. 2017), implemented the boundary conditions to ensure the constant flow rate that matched the capillary number values used in the reference experiments. The protocol suggested here avoids costly

replication of the structure in effort to create a periodic system and does not require a separate imbibition simulation, which is required by the protocol in order to obtain a set of initial configurations with exactly specified saturation levels (Berg, Rücker et al. 2016). Those configurations are needed to explore the relative permeability -- saturation dependence. Setting up a particular volume ratio within the pore space makes the simulations difficult to automate. In our protocol, the saturation levels are pre-set, and simulations are easy to setup.

Flows in the porous media are characterized by the capillary number $Ca = \mu V/\gamma$, which quantifies the relative effect of viscous forces versus capillary forces. Here, μ is the dynamic viscosity, V is the characteristic velocity, and γ is the interfacial tension. The Ohnesorge number $Oh = \mu(\rho\gamma L)^{-1/2}$, where ρ is the fluid density and L is a characteristic length scale, represents the ratio of viscous forces to inertial and capillary forces.

The viscosities of the oil (non-wetting) and water (wetting) phases are set to $\mu_{nw} = 9.2 \cdot 10^{-4} \text{Pa} \cdot \text{s}$ and $\mu_{nw} = 1.0 \cdot 10^{-3} \text{Pa} \cdot \text{s}$, respectively, resulting in a viscosity ratio of $M = 0.92$. During the forced drainage simulation, the capillary number is calculated as: $Ca_{nw} = \mu_{nw} u_{nw}/\gamma = 1.9 \cdot 10^{-3}$, where u_{nw} is the non-wetting phase velocity. Oil is injected until an irreducible water saturation of approximately 80% is achieved, which correlates well with the existing experimental and simulation practices. For the steady state simulation, the inlet velocity is reduced to $u_x^{inlet} = 1.0 \cdot 10^{-5}$ lattice units. The steady state flow was considered achieved when the kinetic energy E_{kin} fluctuated within a small tolerance over successive windows, indicating that the flow properties had stabilized. This methodology ensured that transient fluctuations were minimized and that the system had reached a stable flow regime. The fluid volume exceeding the sample pore volume by the factor of three is pumped through the sample to reach steady-state conditions, which is consistent with the standard core flooding experimental protocol (Busch and Müller 2011). In single-phase simulations, only the oil phase was considered. Absolute permeability κ in single-phase simulations is calculated with the Darcy's law equation in the steady state regime:

$$\kappa = \langle v \rangle \mu \Delta \rho, \quad (3)$$

where $\Delta \rho$ is the effective density difference on the inlet and outlet boundaries, and $\langle v \rangle$ is the average fluid velocity. The permeability is obtained from the LB reduced permeability as $\kappa = k^* s^2$ where s is the resolution of the map. The relative permeability for fluid i in the multiphase flow regime is calculated from the total (absolute) permeability and the flux of the particular phase: $\kappa_{ir} = \kappa_i/\kappa$ and depends on the composition of the fluid in the porous matrix described by the saturation S , which was used as a convergence criterion, particularly considering the presence of immobile water within the pore space. To achieve a target saturation S , we accounted for both immobile water saturation amount of water in the injected mixture. The two-phase injection of oil -- water mixture effectively simulated co-current two-phase flow (Zheng et al. 2021) that is typical to secondary oil recovery via brine filtration.

Single-phase permeabilities of sandstones and carbonates

The samples from Group 1 exhibit the highest porosity and were previously analyzed by Dong (Dong 2008). For the more finely discretized maps of Groups 2, 3, and 4, we computed pore volume distributions (PVDs) and assess pore connectivity and network tortuosity using topological methods from (Schouten and Broek 2014). This approach allows the segmentation of a continuous pore space into discrete effective pores, facilitating the calculation of PVDs while also characterizing connectivity and tortuosity. All Achimov sandstones display unimodal pore volume distributions, with the mode ranging between 420–900 μm^3 (Figure 2), corresponding approximately to a pore radius of 8 μm . The mode of the PVD shows only minor variation with overall porosity. In contrast, the reference samples (Group 2) include both unimodal and bimodal cases (2_BS and 2_KC), where a significant fraction of very small pores has negligible impact on overall permeability. For these samples, the dominant pore volume modes are shifted to values 2–3 orders of

magnitude larger than those of the Achimov group. The Gildehauser sandstone presents a relatively narrow PVD with a distinct peak at a 25 μm pore diameter, alongside the presence of some significantly larger pores (Dong 2008).

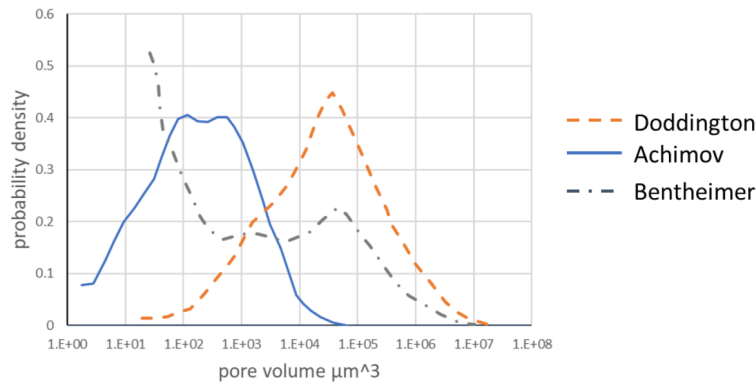


Figure 2—Pore volume distributions for select samples of Groups 2 and 3 in μm^3 . Solid blue line: Achimov sandstone 3_A; Orange dashed line: Doddington sandstone, unimodal distribution; Gray dashed-dotted line: Bentheimer sandstone, bimodal distribution.

Table 1 presents the calculated permeability values alongside experimental data and results from other high-accuracy methods used as modern benchmarks. For wide-pore sandstones, the LBM predictions show good agreement with experimental measurements, with deviations typically within 10% — a range comparable to experimental and modeling errors, and thus sufficient for practical applications. Notably, LBM outperforms PNM in accuracy across all samples, despite PNM's adjustable parameterization.

Table 1—Results of the porosity analysis and permeability calculations with LBM, comparison with literature data.

Sample	Digital map	Resolution, $\mu\text{m}/\text{voxel}$	Effective porosity, %	Mean pore diam, mm	κ , mD (LBM)	κ , mD (lit. data)	Litr.: ref, method,
1_LV60A	450 ³	10	37.7	62.1	30246	35000	exp [p]
1_S1	300 ³	8.683	14.1	53	1525	1678	exp [p]
1_S2	300 ³	4.956	24.6	32	3742	3898	exp [p]
1_S9	300 ³	3.398	22.2	24	2097	2224	exp [p]
1_C1	400 ³	2.85	23.3	17.47	1213	1102	exp [p]
1_A1	300 ³	3.85	42.9	25	6024	7220	exp [p]
2_G	893×897×570	4.92	20	25	1650	1500	exp [p]
2_BS (b)	1000 ³	3	21.6	21.13	2805	3595	DNS [q]
2_DS (c)	1000 ³	2.69	19.5	19.18	3673	3812	DNS [q]
2_EC (d)	1000 ³	3.31	10.9	19.9	313	221.4	DNS [q]
3_A	600 ³	1.2	5.8	3.15	1.14	0.84	DNS [r]
3_B	600 ³	1.2	4.4	5.05	1.24	0.87	DNS [r]
3_C	600 ³	1.2	5.3	3.75	0.87	0.66	DNS [r]
3_D	600 ³	1.2	9.7	5.3	12.9	9.38	DNS [r]
3_E	600 ³	1.2	6.4	4.65	1.69	1.28	DNS [r]
4_A	1400 ³	1.2	8.1	7.15	5.54	3.05	DNS [r]
4_B	1400 ³	1.2	5.1	7.01	3.03	1.6	DNS [r]
4_D	1400 ³	1.2	10.2	7.12	16.3	12.53	DNS [r]

For tight Achimov sandstones (Groups 3 and 4), LBM tends to yield higher permeability values than other direct methods. However, a rigorous experimental validation is lacking for these cases. The DHD and DiMP-hydro results are reasonably consistent with each other, showing no clear systematic deviations. The most significant discrepancies between LBM and alternative methods occur in the tightest, low permeability samples (3_C and 3_E), with the exception of sample 3_B, where LBM and DHD converge closely despite its low permeability. Crucially, in no instance did we observe a severe discrepancy that would undermine LBM's practical utility for permeability prediction.

Samples: (a) Berea sandstone; (b) Bentheimer sandstone; (c) Doddington sandstone; (d) Estailades carbonate; e Ketton carbonate.

Data sources: [p] (Dong and Blunt 2019); [q] (Raeini et al. 2014); [r] (Orlov et al. 2021)

Relative permeabilities in concurrent two-phase flows

The LBM simulations of relative permeabilities in multiphase flows have substantial history, mostly applied to manufactured wide-pore structures (Pan et al. 2004, Ahrenholz et al. 2008, Zacharoudiou and Boek 2016, Zakirov and Khramchenkov 2020). Even fluid morphology was compared with experiments: Zhao et al. 2019 conducted a comprehensive comparison of pore-scale models, including LBM, phase-field, volume-of-fluid, and PNM, using quasi-2D microfluidic flow cells with vertical posts of varying wettability. The quasi-2D structures allowed a detailed study of the fluid configurations and observation of dynamic effects including snap-offs, Haines jumps, and the formation of thin films of the wetting fluid during the fluid displacement process. LB simulations with SC, CG, and FE models all showed a very reasonable agreement with experimental observations. Comparison of relative permeabilities for actual rock samples is quite limited.

Berg et al. (Berg 2014, Berg et al. 2016) investigated steady-state flows in Gildehauser sandstone using the FE LBM method. Their computationally intensive approach involved (i) primary forced drainage and imbibition, (ii) generating a fully periodic pore network by mirroring the initial structure in all dimensions, and (iii) simulating gravitationally driven steady-state flow at fixed fluid volumes. While the predicted permeabilities aligned reasonably with experimental data, the wetting phase exhibited a nearly constant upward shift. Despite its accuracy, the protocol remains cumbersome and resource-intensive. Figure 3 presents the steady-state imbibition relative permeabilities for Gildehauser sandstone obtained via LBM, consistent with ref. (Berg et al. 2016). Overall, the simulations agree well with experiments. All three models (SC, CG, FE) correctly predict immobile water saturation at ~80% and residual oil saturation at 20%. However, the SC model significantly overestimates residual oil saturation, whereas the CG and FE models match it closely. The latter two also accurately reproduce the intersection point of the fluid curves—a key metric for simulation validity.

The SC model's inaccuracies likely stem from spurious velocities at fluid-fluid interfaces. Steady imbibition involves frequent snap-off events, as observed in flow visualizations. Resolving such dynamics demands high spatial resolution, particularly in tight pores, but the available CT scan lacks sufficient detail. Consequently, the SC model overestimates flow velocity and relative permeability.

In contrast, the FE and CG models perform exceptionally well, with the CG model excelling in predicting residual oil saturation and relative permeability curves. This success is attributed to improved wetting boundary conditions, which better capture contact angles and wetting behavior near solid walls—critical for modeling capillary-driven flows. The CG model's explicit incorporation of experimental Oh further enhances its accuracy in simulating tight sandstone multiphase flow. Thus, among the tested models, the CG approach emerges as the most reliable for such applications.

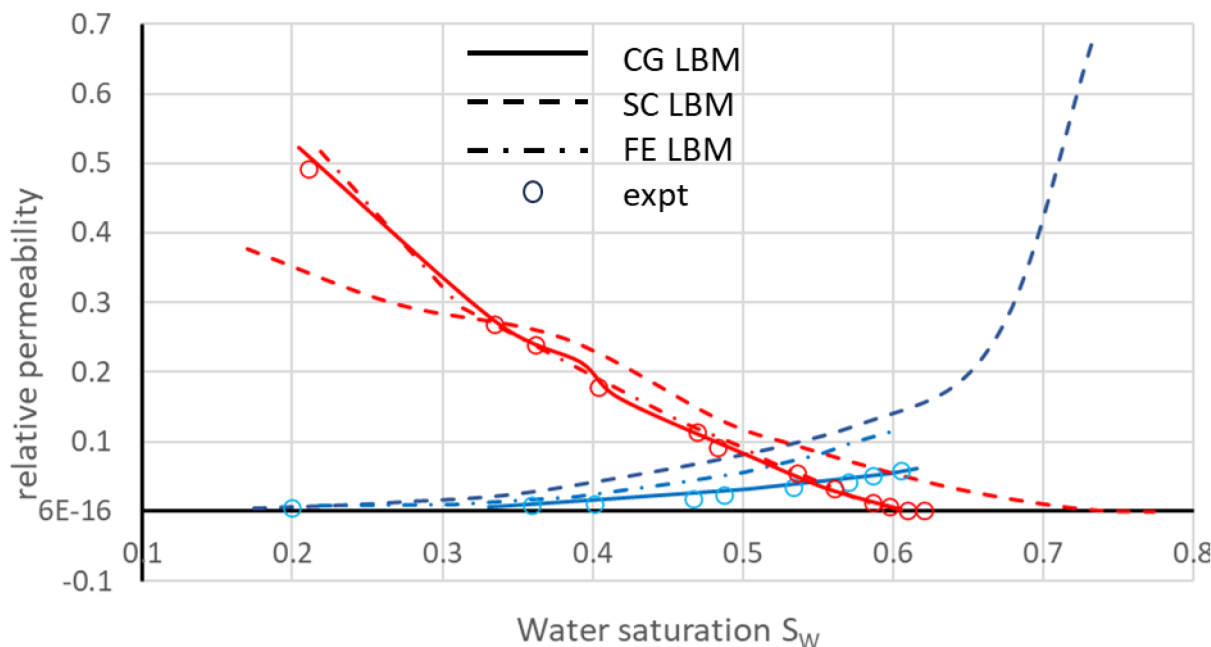


Figure 3—Relative permeability data from steady-state imbibition experiments in Gildehauser sandstone compared with various Lattice Boltzmann simulation approaches. "o" and "w" subscripts denote relative permeabilities of oil and water respectively (full water saturation). Blue: water relative permeability, Red: oil relative permeability. Circles: experimental data, solid line GC LBM, dashed-dotted line: FE LBM, dashed line: SC LBM.

Gas bubble formation and foam drainage

LB simulations of liquid-gas flows are very few, and practically no comparison with experiment has been published. To demonstrate the capability of the approach, we created a foam and simulated the dynamics of foam drainage. To start the drainage simulation, one needs to obtain a foam of the targeted rate and bubble size distribution. This is achieved by placing bubble nuclei in a vessel of the required liquid volume. The gas initially dissolved in the liquid. Thermodynamically, this system is unstable. Note however that the LB fluid cannot undergo spontaneous spinodal decomposition into stable equilibrium liquid and vapor. The finite size nuclei have to be placed, and the gas diffuses into the nuclei, which grow until the Henry law is established. The placement of the nuclei therefore affects the bubble size distribution obtained. The initial nuclei were all $3v_x$ in size; we experimented with regular and random nuclei placement. As the dissolved gas diffuses into the bubbles, they grow. During the growth stage, no gravity is applied. The growth process eventually stops when the concentration of the dissolved gas in the liquid phase reaches the solubility limit. At this moment, marked as $t = 0$, the gravity field is turned on and the foam drainage starts. Foam formation is shown in Figure 4a (nuclei placement) and Figure 4b which shows the foam at $t = 0$. The parameters were fitted to the system explored experimentally in ref. (Maurdev et al. 2006), and a bubble test was conducted as usual.

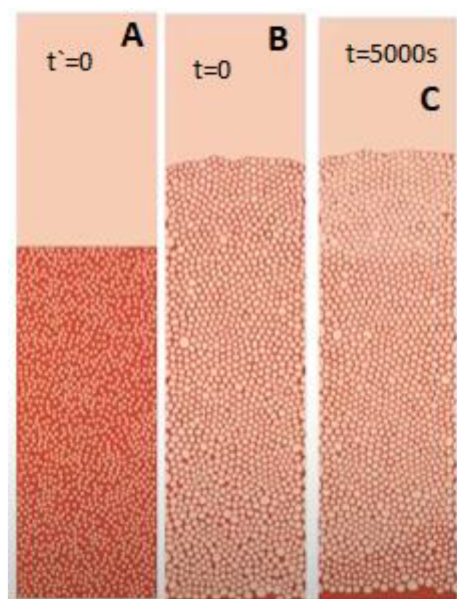


Figure 4—Snapshots of the foam formation and drainage simulation. (A) random bubbles are initiated and allowed to grow until a foam forms (B) gravity field turned on (C) liquid flows down and foam rate decreases as the depth increases.

Figure 5 shows the dependence of the height of the water layer at the bottom of the vessel on time. It is clear that the model reasonable reproduces the drainage process under the gravity field. Almost ideal agreement with the experiment was obtained with a randomized bubble placement, but quantitative conclusion is difficult at this time.

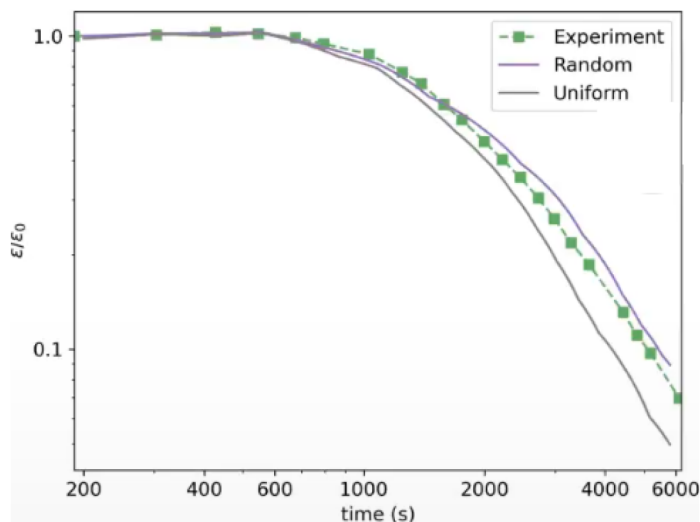


Figure 5—Dependence of the foam rate on time for different layers: experiment, LB simulation with regular/uniform and random nuclei placement on foam formation

Conclusion

The lattice Boltzmann (LB) method employing multiple relaxation times and half-bounceback boundary conditions demonstrates good quantitative accuracy in predicting permeabilities, even for tight low-porosity sandstones. For wide-pore samples, LB results show slightly better agreement with experimental data compared to state-of-the-art direct Navier-Stokes solvers. In tight-pore formations, the method maintains satisfactory consistency with other advanced computational approaches - the observed discrepancies

between LB and alternative direct methods remain comparable to variations among published simulation results [REFS].

Our analysis reveals two characteristic trends in LB permeability predictions:

A moderate overestimation for extremely tight, low-permeability samples, attributable to reduced fluid-wall friction representation when using half-bounce-back boundary conditions;

A slight underestimation for more permeable, open-porosity samples, resulting from neglected compressibility effects.

For multiphase flows, this study introduces an optimized protocol for steady-state simulations in porous media, closely mimicking experimental concurrent flow conditions. The procedure consists of two key steps: (i) initial drainage to achieve 80% non-wetting fluid saturation, followed by (ii) a sequence of steady-state two-phase flow simulations with controlled inflow fluid ratios. This approach directly generates the relative permeability-saturation relationship while offering significant computational advantages over current best practices—eliminating the need for system periodization via map replication and bypassing separate imbibition simulations to establish target saturations.

We demonstrate this protocol through two-phase flow simulations in a digital twin of Gildehauser sandstone, employing three distinct LBM interfacial interaction models. Although the sample possesses a relatively simple pore structure, the flow dynamics involve challenging phenomena like frequent droplet snap-offs. Among the tested approaches:

The **SC model** provides computational efficiency and versatility by avoiding explicit interface tracking, though it demands high spatial resolution to minimize spurious velocities during interfacial events.

The **CG model** explicitly resolves fluid interfaces, enabling precise interfacial dynamics representation and achieving near-perfect agreement with experimental permeability-saturation data.

The **FE model** incorporates a free-energy functional to consistently model interfacial tension effects.

While the CG model delivers quantitatively accurate results, the SC model systematically overestimates permeabilities—a consequence of its inherent resolution requirements and lack of explicit interface tracking, which leads to velocity artifacts during snap-off events. These findings highlight the critical balance between computational efficiency and physical accuracy in pore-scale multiphase simulations.

To our knowledge, these gas-liquid flow simulations represent the first comprehensive quantitative validation of the lattice Boltzmann method (LBM) beyond elementary benchmark tests (e.g., bubble tests). The implemented LB framework incorporated all essential physical mechanisms: vapor-liquid equilibrium, gas transport in the liquid phase, disjoining pressure effects. The disjoining pressure parameters in this work were not specifically calibrated to experimental data.

When validated against foam drainage experiments, the LBM demonstrated good agreement with measured results, though the simulations showed notable sensitivity to initial conditions. These findings support the conclusion that LBM serves as a robust and versatile computational tool for pore-scale simulation of diverse upstream processes involving multiphase flows, including: filtration processes, acid stimulation treatments, and condensate accumulation during gas production.

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